```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
uploaded = files.upload()

# Get uploaded image name
filename = list(uploaded.keys())[0]

# Read image
image = cv2.imread(filename)

# Check image
if image is None:
    print("Error: Image could not be loaded.")
else:
    # Laplacian mask
    kernel = np.array([
        [0, 1, 0],
        [1, -8, 1],
        [0, 1, 0]
    ], dtype=np.float32)

    # Apply filter
    laplacian = cv2.filter2D(image, cv2.CV_32F, kernel)

    # Sharpen image
    sharpened = image.astype(np.float32) - laplacian

    # Keep pixel values between 0 and 255
    sharpened = np.clip(sharpened, 0, 255).astype(np.uint8)

    # Convert BGR to RGB for displaying
    image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
    sharpened_rgb = cv2.cvtColor(sharpened, cv2.COLOR_BGR2RGB)
```

```
# Display images
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(sharpened_rgb)
plt.title("Sharpened Image")
plt.axis("off")

plt.show()

# Save sharpened image
cv2.imwrite("Sharpened.jpg", sharpened)

print("Sharpened image saved successfully!")
```

Choose Files Screenshot… 091603.png

Screenshot 2026-08-19 091603.png(image/png) - 902271 bytes, last modified: 8/19/2026 - 100% done

```
Saving Screenshot 2026-08-19 091603.png to Screenshot 2026-08-19 091603.png
```

Original Image

Sharpened Image

```
Sharpened image saved successfully!
```